

Ground-Based Cloud Detection Using Automatic Graph Cut

Shuang Liu, Zhong Zhang, Baihua Xiao, and Xiaozhong Cao

Abstract—Ground-based cloud detection plays an essential role in meteorological research, and object segmentation techniques have recently been introduced to solve this issue. As a kind of object segmentation technique, interactive graph cut has emerged as a very powerful tool due to its effective segmentation ability. However, it requires users to provide labels for certain pixels as “object” or “background,” which inevitably prohibits automatic cloud detection in large-scale applications. In this letter, we focus on the issue of automatic cloud detection and propose a novel algorithm named as automatic graph cut. We treat clouds as a special kind of object and eliminate human labeling by two procedures. First, we adaptively compute the thresholds for each cloud image which automatically label some pixels as “cloud” or “clear sky” with high confidence. Then, those labeled pixels serve as hard constraint seeds for the following graph cut algorithm. The experimental results show that the proposed algorithm not only achieves better results than the state-of-the-art cloud detection algorithms but also achieves comparable results with the interactive segmentation algorithm.

Index Terms—Automatic graph cut (AGC), ground-based cloud detection.

I. INTRODUCTION

CLOUDS, which are one of the most important meteorological phenomena related to the hydrological cycle, strongly affect the Earth’s energy balance. Most of existing cloud-related research requires the technology of ground-based cloud observation, such as ground-based cloud classification and cover evaluation. The successful evaluation of cloud cover can improve the precision of weather prediction and understanding of climatic conditions. Moreover, cloud cover evaluation also plays a guiding role in flight planning and aviation [1]. The target of cloud cover evaluation is to estimate the proportion of cloud pixels among all pixels in a ground-based cloud image. At present, cloud cover evaluation is mainly conducted by professionally trained observers who manually estimate the



Fig. 1. Examples of ground-based cloud images.

coverage [2], [3]. However, this method relies heavily on expert domain knowledge and extensive human efforts. Moreover, it might suffer from classification ambiguities due to the diversity of multiple observers. Therefore, automatic ground-based cloud cover evaluation is highly desired in this area. Cloud detection, which classifies each pixel into cloud or clear-sky element, is a fundamental task for cloud cover evaluation. In this letter, we focus on the cloud detection. However, this issue is particularly challenging due to the severe illumination changes and vague boundaries between cloud and sky regions, as shown in Fig. 1. From this figure, we can see that these three cloud images possess significant different illumination conditions, and the red rectangles indicate the vague boundaries.

To address the cloud detection, a lot of ground-based imaging devices have been developed for capturing cloud images, such as the whole sky imager [4], [5], total sky imager [6], [7], whole sky camera [7], infrared cloud imager [8], and all-sky imager [9]. Based on these devices, a large number of methods have been proposed for ground-based cloud detection. Due to the scattering difference between cloud particles and air molecules, most cloud detection algorithms treat color as the primary feature to distinguish cloud from clear sky. Long *et al.* [7] proposed a threshold algorithm for cloud detection according to a certain ratio of R over B intensity using a red–green–blue (RGB) image. Specifically, pixels (except the particular case such as pixels in the horizon or around the sun) with R/B values greater than 0.6 are identified as cloud; otherwise, they are identified as clear sky. Kreuter *et al.* [11] reported a different threshold of 0.77 on the R/B for identifying clouds, which is illustrated to be a more suitable choice than the previously mentioned value of 0.6. Other algorithms are proposed to classify the cloud based on other color features, such as saturation [12], difference value [13], and the Euclidean geometric distance [14]. This kind of methods can be referred to as the fixed threshold algorithm. However, fixed threshold methods are unsuitable for different sky conditions. To overcome this drawback, adaptive threshold algorithms are investigated in [2]. Moreover, neural network model [9] and superpixel segmentation [10] are also applied for cloud detection. Although certain improvement can

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be achieved, their performance is still unsatisfactory for real-world applications.

Identifying clouds is the goal of sky imaging algorithms, and object segmentation techniques have recently been introduced to solve this issue. In this letter, we propose a novel segmentation algorithm named automatic graph cut (AGC) for cloud detection. We first adaptively compute the thresholds for each cloud image which automatically label some pixels as cloud or clear sky with high confidence. Then, those labeled pixels serve as hard constraint seeds for the following graph cut algorithm. The experimental results show that the proposed algorithm not only achieves better results than the state-of-the-art cloud detection algorithms but also achieves comparable results with the interactive segmentation algorithm.

The remainder of this letter is organized as follows. Section II details the proposed algorithm. The experimental results and comparative evaluation are given in Section III. Finally, we conclude this letter in Section IV.

II. PROPOSED ALGORITHM

Given a ground-based cloud image, our goal is to extract the cloud pixels (foreground) from the clear-sky pixels (background) precisely. The interactive graph cut is an effective algorithm for segmentation. However, one drawback is that the interactive graph cut needs users to manually label hard constraints by assigning several pixels (seeds) to foreground and background. In this letter, a novel cloud detection algorithm based on AGC is proposed. The proposed method consists of two stages: 1) According to the characteristics of ground-based cloud images, we automatically label some pixels with high confidence as “cloud” or “clear sky” elements, and 2) those labeled pixels serve as hard constraint seeds for the subsequent graph cut algorithm. The details of the proposed AGC algorithm are given hereinafter.

A. Automatically Acquiring Hard Constraints

Since the ground-based cloud image is obtained from outdoor environment, the large illumination variations are inevitable. Moreover, vague boundaries between cloud and sky regions as the intrinsic characteristics of the cloud image are ubiquitous. The aforementioned characteristics make cloud detection challenging. In this letter, we propose to automatically assign several pixels with high confidence as hard constraint seeds, which form part of the foreground and background. As stated in Section I, color features are beneficial to distinguishing cloud from clear sky. In addition, adaptive threshold algorithms are suitable for different weather conditions in some extent [2]. Based on these properties, the hard constraint seeds are obtained as follows.

- 1) An RGB cloud image is transformed into a single-channel feature image. We first use the ratio R/B as the feature image and then normalize the elements of the R/B image to the range of 0–255, denoting the feature image as I_f . In this way, the difference between cloud and clear-sky elements is increased, and illumination variations are alleviated in some extent.

- 2) The threshold T is computed on the feature image using the Otsu algorithm [15]. It is based on the different distribution of cloud and clear-sky elements in a cloud image, i.e., the distinct bimodal distribution in the histogram derived from the cloud image. We need to find a value which can maximize the variance between cloud and clear-sky elements. It can be formulated as

$$V(T) = P_C P_S (\mu_C - \mu_S). \quad (1)$$

P_C and P_S are the occurrence probabilities of the cloud and clear-sky elements, respectively, which are computed by $P_C = N_C/N_T$ and $P_S = N_S/N_T$. For a cloud image, N_C denotes the number of cloud pixels, N_S denotes the number of clear-sky pixels, and N_T denotes the total number of pixels. μ_C and μ_S are the mean values of cloud and clear sky, respectively. Finally, T is set to be $\max\{V(T)\}$ ($1 < T < 255$).

- 3) If we directly employ T as the threshold, then some pixels will be misclassified. In order to obtain hard constraint seeds with high confidence, we regulate the threshold T in the R/B feature space. Concretely, pixel values greater than θT are identified as cloud elements, and the pixel values less than $(1/\theta)T$ are identified as clear-sky elements. It can be formulated as the following:

$$\begin{cases} i \in \text{cloud pixels}, & I_f^i > \theta T \\ i \in \text{clear sky pixels}, & I_f^i < (1/\theta)T. \end{cases} \quad (2)$$

Here, i is the pixel of the cloud image; θ is a parameter controlling the confidence of hard constraint seeds, where its value is greater than 1.

The strategy of automatically labeling hard constraint seeds is feasible in practice since we use a parameter θ to regulate the threshold T , which guarantees the stability of assigning hard constraint seeds. Since the threshold θT is high enough, it can ensure that pixels with an R/B feature larger than θT are cloud elements. Likewise, the threshold $(1/\theta)T$ is low enough, so it can also ensure that pixels with an R/B feature less than $(1/\theta)T$ are clear-sky elements. In addition, this form of θT (or $(1/\theta)T$) is invariant to the monotonic change of illumination. We visualize hard constraint seeds to verify the reliability of automatically labeling seeds. Fig. 2 shows the examples of automatically labeling hard constraint seeds in cloud images. The pixels with green are the seeds of cloud elements, and the pixels with red are the seeds of clear-sky elements. We empirically set $\theta = 1.3$, and the value is invariable in the following experiments. From Fig. 2, we can see that these seeds belong to their truth classes (cloud or clear sky).

B. Graph Cut Algorithm

After obtaining the hard constraint seeds, we conduct the graph cut algorithm. Let $G = \{V, E\}$ be a weighted undirected graph, where V is a set of nodes (the pixels of the cloud image) and E denotes a set of edges connecting two nodes. Each edge in the graph is assigned to a nonnegative weight w_e (cost). There are also two special nodes named terminals:

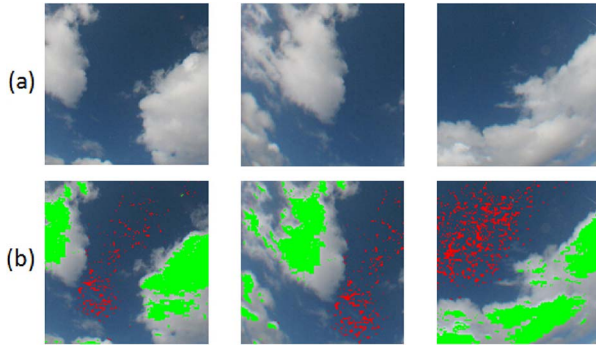


Fig. 2. Examples of automatically labeling hard constraint seeds in the cloud images. The pixels with green are the seeds of cloud elements, and the pixels with red are the seeds of clear-sky elements.

the foreground and background terminals. A cut is a subset of edges so that each node is assigned to either of the terminals. The cost of a cut is usually defined as the sum of the costs of edges [16]

$$|C| = \sum_{e \in C} w_e. \quad (3)$$

The cost function that we use as the soft constraints for segmentation should include region and boundary properties. Let P denote all the pixels in the image and N denote the set of pairs (p, q) where p and q are the neighboring pixels in P . $L = \{L_1, L_2, \dots, L_p, \dots\}$ is a binary vector whose components L_p specify the labels of pixel p in P . Each L_p could be either 1 (foreground) or 0 (background). The cost function $E(L)$ for each segmentation is defined as

$$E(L) = \lambda \cdot R(L) + B(L) \quad (4)$$

where

$$R(L) = \sum_{p \in P} R_p(L_p) \quad (5)$$

$$B(L) = \sum_{(p,q) \in N} B_{(p,q)} \cdot \delta(L_p, L_q) \quad (6)$$

$$\delta(L_p, L_q) = \begin{cases} 1, & \text{if } L_p \neq L_q \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

The coefficient $\lambda \geq 0$ in (4) specifies the relative importance between the region cost $R(L)$ and the boundary cost $B(L)$. The region cost $R(L)$ measures the individual penalty for labeling the pixel p as the foreground or background. Each pixel has two cost weights $R_p(1)$ and $R_p(0)$ corresponding to the case of connecting with the foreground and background, respectively. The boundary cost $B(L)$ reflects the penalties for the discontinuity between neighboring pixels $\{p, q\}$. $B_{(p,q)}$ should be large when the values of pixels p and q are similar, and $B_{(p,q)}$ is close to zero when the two pixel values are very different.

In this letter, $R(L)$ and $B(L)$ in (4) are given in the following steps.

- 1) Assume that we get n foreground seeds and m background seeds denoted as $\{Fore\}_n$ and $\{Back\}_m$, respectively.

TABLE I
AVERAGE DETECTION ACCURACY OF DIFFERENT ALGORITHMS

	Rad	Diff	Adp	Ours
precision(%)	72.53	73.46	83.32	94.79

- 2) We calculate the color feature distance between each pixel p and each seed, denoted as $D\{Fore\}_n^p$ and $D\{Back\}_m^p$, respectively.
- 3) $R_p(1)$ and $R_p(0)$ can be calculated as follows:

$$R_p(1) = \min \{D\{Fore\}_k^p\}, k = 1, 2, \dots, n \quad (8)$$

$$R_p(0) = \min \{D\{Back\}_k^p\}, k = 1, 2, \dots, m. \quad (9)$$

- 4) The boundary cost $B(L)$ is a decreasing function of the distance between pixels p and q . For a cloud image, we use the function

$$B_{\{p,q\}} = \exp\left(-\frac{(f_p - f_q)^2}{2\sigma^2}\right) \quad (10)$$

where f_p and f_q denote the feature vectors and their values are R/B . σ is a factor which is set to 0.3 in this letter.

To obtain the values of all the parameters, we consider how to solve this segmentation problem. The min-cut/max-flow algorithm in [17] is applied for solving (4), and Boykov and Jolly [16] proved that this min-cut algorithm would give a segmentation minimizing among all the segmentation.

C. Analysis

The aforementioned strategy can effectively detect clouds when clouds and clear sky appear at one time. Nevertheless, there is an extreme case that the images are fully covered by clouds or clear sky. The proposed AGC can also provide a guiding role for this case. When a cloud image is fully covered by clouds or clear sky, the variance of pixel intensities is small. Hence, almost none of pixel intensities is larger than θT (or smaller than $(1/\theta)T$). In this case, we calculate the average value R_{ave} in the original R-channel image. If $R_{ave} < 120$, we classify the image to clear sky; otherwise, the cloud image is classified as cloudy image. It is because the image with clear sky has a smaller R-channel intensity. In a word, when given a cloud image I , the flowchart of our algorithm is accomplished by the following steps.

- 1) Transfer the RGB image to three-channel images (R-channel image, G-channel image, and B-channel image), and then, use the R-channel/B-channel as the R/B feature image.
- 2) Normalize the intensities of the R/B feature image to the range of 0–255, denoting this kind of feature image as I_f .
- 3) Calculate the threshold T for image I_f .
- 4) Calculate θT and $(1/\theta)T$ to regularize the threshold T .
- 5) If there are no pixels larger than θT or smaller than $(1/\theta)T$, then calculate the average value R_{ave} of the original R-channel image in I .
 - a) If $R_{ave} < 120$, image I is clear sky, and go to step 8).
 - b) If $R_{ave} \geq 120$, image I is cloudy, and go to step 8).

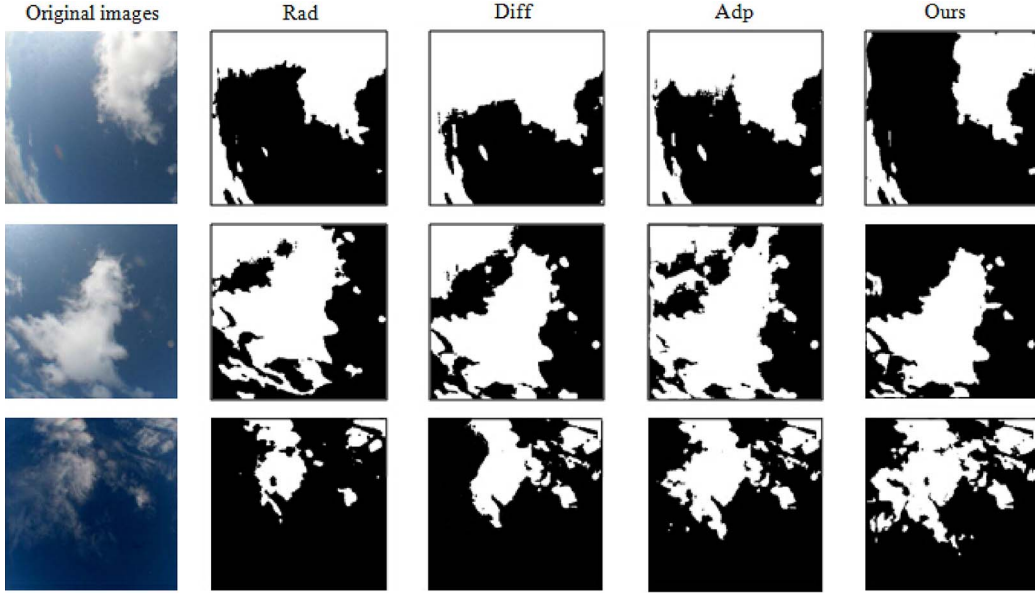


Fig. 3. Experimental results of four different detection algorithms.

- 6) If step 5) does not satisfy, pixels larger than θT serve as the cloud's hard constraint seeds, and pixels smaller than $(1/\theta)T$ serve as the clear skies' hard constraint seeds.
- 7) Input these hard constraint seeds into the graph model, and then, go to step 8).
- 8) Return the final detection results.

III. EXPERIMENTAL RESULTS

A. Database

The equipment used to capture the ground-based cloud images is a digital camera equipped with a fisheye lens, which provides a field of view larger than 180° . Its aperture is f/1.2, exposure time is 1/200, ISO is 100, and focal length is 8 mm. The camera is set to capture one cloud image per 15 s. More information about the camera can be found in [18]. In this letter, we test our AGC on the database of such images, which is captured during a transit of the German research vessel “Polarstern” from Germany to South Africa. In the course of this trip, different climate zones and different seasons were crossed, and therefore, the data covered a wide range of sky conditions and solar zenith angles. The square area of an image is cropped as shown in Fig. 1. Then, each image is resized to 200×200 pixels. The database consists of 300 cloud images.

B. Methods in Comparison Study

In this letter, the proposed method is compared with three current cloud detection methods.

- 1) Ratio threshold (Rad) [7]: We assign $R/B = 0.6$ as the threshold. The pixels with $R/B > 0.6$ are classified as cloud; otherwise, they are classified as clear sky.
- 2) Difference threshold (Diff) [12]: We assign $R - B = 30$ as the threshold. The pixels with $R - B > 30$ are classified as cloud; otherwise, they are classified as clear sky.

- 3) Adaptive threshold (Adp) [2]: The threshold is calculated by the Otsu algorithm [15].

C. Experimental Results and Analysis

To evaluate the effectiveness of the proposed algorithm for the cloud detection, we carry out a series of experiments on the database mentioned in Section III-A. Furthermore, meteorological experts manually segment the 300 cloud images into binary masks as the ground truth. Quantitative evaluations of all the aforementioned algorithms are performed as

$$\frac{NT_{cloud} + NT_{sky}}{N} \quad (11)$$

where NT_{cloud} and NT_{sky} are the numbers of the true cloud pixels and true clear-sky pixels, respectively. N is the total number of pixels in the ground-based image. Table I lists the detection performance of four algorithms, which is the average of the accuracy for all 300 test images. We can see that our method achieves the highest detection accuracy of 94.79%, which is significantly higher than that of other algorithms. Fig. 3 shows the results of some examples detected by the four algorithms, where several conclusions can be drawn. First, the “Rad” and “Diff” algorithms are not suitable for different weather conditions. The threshold is too large for the first and second rows, which results in mislabeling clear-sky elements as clouds. On the other hand, the threshold is too small for the third row, which leads to mislabeling clouds as clear sky. Second, the “Adp” algorithm cannot solve the conditions with nonuniform illumination shown in the first row. Third, the proposed algorithm obtains the best detection results in all weather conditions, due to assigning pixels with high confidence as the hard constraint seeds.

Fig. 4 shows the detection results of images in the extreme case. (a) and (b) denote the detection results of images fully covered by clouds, and (c) and (d) denote the detection results

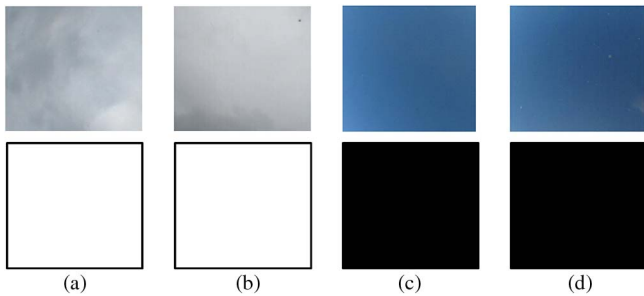


Fig. 4. (a) and (b) denote the detection results with images fully covered by clouds. (c) and (d) denote the detection results with image fully covered by clear sky.

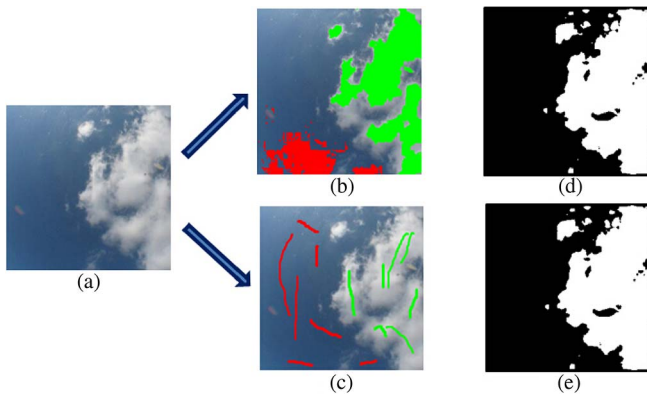


Fig. 5. (a) Original cloud image. (b) Labeling hard constraint seeds automatically. (c) Labeling hard constraint seeds manually. (d) Proposed AGC detection results. (e) Interactive graph cut detection results.

of images fully covered by clear sky. For the four examples, the average values R_{ave} in the original R-channel image are 55.67, 57.83, 182.54, and 169.36, respectively, which proves that setting the threshold of 120 is reasonable.

Finally, we also compare the proposed algorithm with the interactive graph cut. We select 30 cloud images and record the computational time of the two methods. The results show that the average computational time of the AGC algorithm and interactive graph cut method is 4.6 and 12.4 s, respectively. Since the interactive graph cut needs users to label hard constraint seeds manually, it is time consuming. Clearly, the cost of our AGC algorithm is much less. Fig. 5 shows the detection results. We can see that the proposed algorithm achieves comparable detection results with the interactive graph cut. Obviously, our algorithm overcomes the shortcomings arising from manually labeling hard constraint seeds.

IV. CONCLUSION

In this letter, we have proposed a novel algorithm based on AGC to segment ground-based cloud images. This model could automatically label pixels with high confidence as “cloud” or

“clear sky,” and those labeled pixels serve as hard constraint seeds for the following graph cut algorithm. The experimental results show that the proposed algorithm not only achieves better results than the state-of-the-art cloud detection algorithms but also achieves comparable results with the interactive segmentation algorithm.

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